Table 1: IRT criterion of the triangulation ranking. Each item's trace plot was scored from 0 (poor) to 3 (good) for monotonicity (Mon.), discrimination (Disc.), middle-category intersection and non-redundancy (Mid.) and latent-trait coverage (Cov.). Total is the sum of the four scores (0–12). The IRT rank (1 = best) was agreed by the team after inspecting the plots together with these scores, so it does not always follow the total. Within each factor, items are ordered by IRT rank; – = not ranked.

Code	Item	Mon.	Disc.	Mid.	Cov.	Total	IRT rank
Task_cognition Factor							
P041	I found the system's components worked seamlessly together	3	2	2	2	9	1
P060	I enjoyed using this system	2	2	2	3	9	2
P064	The tool encouraged me to think deeply about my data	3	2	2	3	10	3
P062	The system encouraged me to explore my data further	3	3	2	3	11	4
P027	I could complete my essential tasks efficiently	2	3	2	2	9	5
P072	I felt satisfied with this system	1	3	0	2	6	6
P208	I found the system interesting to use	2	2	2	2	8	7
P083	The system stimulated my thinking process	2	2	2	3	9	8
P029	The system provided useful information to help me accomplish my tasks	2	3	2	2	9	–
P052	The system helped me to focus on my tasks	2	3	2	2	9	–
P030	The system was effective for my tasks	2	3	1	1	7	–
P057	The system helped me to gain knowledge about my data	2	3	1	1	7	–
P062C	The system helped to form and evaluate hypotheses about my data	2	3	1	1	7	–
P037	The system integrates well into my overall data analysis workflow	2	2	1	1	6	–
P301	The system was useful for me	2	2	1	1	6	–
P302	The system helped me to understand the data	2	2	1	1	6	–
P025	Once I knew how to use it, the tool made it easy to get answers to my questions about the data	2	2	1	0	5	–
P04A	I would use this system frequently if I had a reason to	0	2	0	2	4	–
P4026B	Once I learned to use it, accomplishing my tasks was straightforward	1	2	0	0	3	–
System Factor							
P214	Using this system did not require a lot of effort from me	3	3	3	3	12	1
P088	I found the system simple to understand	2	3	3	3	11	2
P024	Using the tool's options and controls was intuitive for me	3	3	3	3	12	3
P4026A	Getting the tool configured for my analysis was straightforward	3	3	3	3	12	4
P026	I found it easy to get started using this system	3	2	3	3	11	5
P086	The system was self-explanatory	3	3	3	3	12	6
P074	I did not feel frustrated when using this system	2	3	3	3	11	7
P054	I found it intuitive to interact with the system	2	3	2	3	10	8
P204	I only needed to learn a few things before I could get going with this system	3	2	1	3	9	9
P074A	I did not feel mentally drained after using this system	3	2	3	3	11	10
P022	I can easily remember how to use the system	2	2	2	3	9	11
P203	When I made a mistake, it was easy to get back on track	2	2	2	3	9	15
P084	I felt confident that I was using the system correctly	2	3	2	3	10	17
P096	I found it straightforward to link information across different visualizations in the system	2	2	2	3	9	18

Code	Item	Mon.	Disc.	Mid.	Cov.	Total	IRT rank
Visualization Factor							
P097	Using the system, I could easily retrieve information from my data	1	3	2	0	6	1
P022A	I can easily remember how to read the visualizations in the system	1	3	2	2	8	2
P045A	The system reacted to my inputs as I expected	2	2	3	2	9	3
P068	I found the system consistent in that the same actions led to the same outcomes	2	3	2	2	9	4
P067	I could safely engage in a trial-and-error analysis process	1	2	2	2	7	5
P097A	I could easily navigate between visualizations	1	2	1	2	6	6
P098	I found the visual representations provided by the system were appropriate	1	2	2	3	8	7
P089	The visual representations I saw were simple to understand	1	2	1	1	5	8
P012	The system felt consistent while I was interacting with the data	2	2	1	1	6	9
P033	I found the system's response time satisfactory	1	2	1	3	7	10
P106	The visualizations I saw seemed to represent the data faithfully	2	2	1	0	5	11
P014	I could complete my analysis without having to switch to another tool	0	0	2	3	5	12
P05	The visual representations I saw were not too complex for me	1	2	1	3	7	13
P206	I could easily identify relevant information in the visualizations provided by the system	1	3	0	1	5	14
P211	The system made it easy to visualize my data in different ways	1	2	0	1	4	15
P205	I could clearly see data features (for example, a minimum, an outlier, or a trend) in the visualizations provided	1	2	0	0	3	16